

Software Requirements Specification (SRS)

AI Resume Analyzer

➤ Introduction

1.1 Purpose

The purpose of the AI Resume Analyzer is to help job seekers improve their resumes by comparing them with a job description. The system analyzes resumes using Artificial Intelligence and provides an ATS (Applicant Tracking System) score, identifies missing keywords, suggests improvements, and recommends skills to increase the chances of getting shortlisted.

1.2 Scope

The application allows users to:

- Upload resumes (PDF/DOCX).
 - Enter or upload a job description.
 - Analyze the resume using AI.
 - Generate an ATS compatibility score.
 - Highlight missing keywords and skills.
 - Provide grammar and formatting suggestions.
 - Download the analysis report.
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2.1 Product Perspective

- The AI Resume Analyzer is a web-based application that uses AI and Natural Language
- Processing (NLP) to compare resumes with job descriptions.

2.2 Product Functions

- User registration and login.
- Resume upload.
- Job description input.
- AI-based resume analysis.
- ATS score generation.
- Keyword matching.
- Skill gap analysis.
- Resume improvement suggestions.
- Downloadable analysis report.
- Admin dashboard for managing users and reports.

2.3 User Classes

User (Job Seeker): Uploads resumes and views analysis.

Admin: Manages users, system settings, and reports.

3. Functional Requirements

- User Registration and Login.
 - Upload Resume (PDF/DOCX).
 - Enter Job Description.
 - AI Resume Parsing.
 - Resume and Job Description Comparison.
 - ATS Score Calculation.
 - Missing Keyword Detection.
 - Skill Recommendation.
 - Grammar and Formatting Suggestions.
 - Report Generation (PDF).
 - User Profile Management.
 - Admin Dashboard.
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4. Non-Functional Requirements

- **Performance:** Analysis should complete within 10 seconds.
 - **Security:** Passwords must be encrypted and user data protected.
 - **Reliability:** System availability should be at least 99%.
 - **Usability:** Simple and user-friendly interface.
 - **Scalability:** Support multiple users simultaneously.
 - **Compatibility:** Works on Chrome, Edge, Firefox, and mobile browsers.
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5. Software Requirements

- Frontend: HTML, CSS, JavaScript, React.js
- Backend: Node.js / Express.js
- Database: MongoDB

- AI/NLP: Python, spaCy, OpenAI API (optional)
- PDF Parser: pdf-parse or PyPDF2

7. System Features

- Secure Login
 - Resume Upload
 - Job Description Input
 - AI Resume Analysis
 - ATS Score Display
 - Keyword Matching
 - Skill Gap Analysis
 - Resume Suggestions
 - Download Analysis Report
 - Admin Management
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8 Constraints

- Internet connection is required.
 - Resume must be in PDF or DOCX format.
 - AI analysis accuracy depends on the quality of the uploaded resume and job description.
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9. Assumptions

- Users upload valid resumes.
 - Job descriptions are complete and relevant.
 - AI service is available during analysis.
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10. Future Enhancements

- Interview question prediction.

- Resume builder.
 - Cover letter generator.
 - Job recommendation system.
 - Multi-language resume analysis.
 - Mobile application support
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➤ **Conclusion**

The **AI Resume Analyzer** is an intelligent and user-friendly web application designed to help job seekers improve their resumes using Artificial Intelligence (AI) and Natural Language Processing (NLP). The system analyzes uploaded resumes, compares them with